

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 12:07 UTC

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Generated 2026-05-21 12:07 UTC

- Cycles recorded: **1020**
- Time span: 662.4h (~1.54 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	973
BARRIER_HIT	27
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	59
Schur-Weyl Duality	36
Free Probability	26
Representation Theory of Symmetric Groups	24
Matroid Theory	23
Free Probability Theory	19
Additive Combinatorics	15
Noncommutative L^p Geometry	12
Random Matrix Theory	11
Spectral Graph Theory	11
Noncommutative Harmonic Analysis	10
Polymatroid Theory	9
REAL_ALGEBRAIC_GEOMETRY	9
Algebraic Geometry	8
Persistent Homology	8
Algebraic Topology	7
Finite Geometry	6
Schur-Weyl duality	6
Plethysm Theory	6
Invariant Theory	6
Fourier Analysis on Boolean Functions	5
Additive combinatorics	5

Field	Cycles
Algebraic Geometry of Secant Varieties	5
FOURIER_ANALYSIS	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Non-Abelian Harmonic Analysis	4
COMMUNICATION_COMPLEXITY	4

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-21 09:59 UTC	INCONCLUSIVE	Discrete Morse Number of Rejection Complex Bounds
2026-05-21 10:06 UTC	INCONCLUSIVE	ACC ⁰ [m] Size
2026-05-21 10:14 UTC	INCONCLUSIVE	Trace-Norm Capacity
2026-05-21 10:22 UTC	INCONCLUSIVE	Conjecture for Read-Twice BP
2026-05-21 10:32 UTC	INCONCLUSIVE	Communication Ma
2026-05-21 10:48 UTC	INCONCLUSIVE	Slice-Fourier Variance
2026-05-21 10:54 UTC	INCONCLUSIVE	Asymmetry of Perm vs
2026-05-21 11:03 UTC	INCONCLUSIVE	Monotone Det Substitu
2026-05-21 11:12 UTC	INCONCLUSIVE	VC dimension of gadget row
2026-05-21 11:23 UTC	INCONCLUSIVE	family tightly bounds lifted
2026-05-21 11:29 UTC	INCONCLUSIVE	D ^{cc}
2026-05-21 11:40 UTC	INCONCLUSIVE	Gromov 4-Point Hyperbolicity
2026-05-21 11:54 UTC	INCONCLUSIVE	Gap Separates k-CLIQUE
		Monotone DNFs
		Lorentzian Hessian Gap of
		Spanning-Tree Polynomial
		Bounds Max-CUT
		NW-Design Slice Spectral
		Discrepancy Bound for DISJ
		MST Persistence Entropy
		Lower-Bounds Log-Rank of
		Sign Matrices
		Instance-Complexity
		Certification Depth Bounds
		Monotone DNF Size
		Cyclic Fourier Spread of
		Term-Vertex Incidence in
		Monotone k-CLIQ
		Coboundary Spectral Gap
		Lower-Bounds DPLL Tree Size
		on Tseitin
		Heilmann-Lieb Matching Root
		Gap Bounds Monotone
		k-CLIQUE DNF Size
		Fourier L1 Norm
		Cubed-Polynomial Bound for
		Read-Twice BPs

Time	Verdict	Title
2026-05-21 12:01 UTC	INCONCLUSIVE	Apolar Catalecticant Defect Bounds DISJ Randomized Communication
2026-05-21 12:07 UTC	BARRIER_HIT	A_5 Barrington-Hash Equidistribution Gap for DISJ

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```